

Factory Reallocation & Shipping Optimization Recommendation System for Nassau Candy Distributor Using Machine Learning

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Abstract—This project presents a Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor using Machine Learning techniques. Traditional factory allocation methods rely on fixed rules that often increase shipping costs, delivery time, and operational expenses. The proposed system analyzes historical sales, product information, shipping mode, region, and profit data to identify the most efficient factory allocation strategy. Three machine learning algorithms—Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor—were implemented and evaluated using MAE, MSE, RMSE, and R^2 Score. Random Forest achieved the highest prediction accuracy and was selected as the final model. A Streamlit dashboard was also developed to provide interactive business insights, factory recommendations, and data visualizations for decision-making.

Keywords—Machine Learning, Random Forest, Factory Allocation, Supply Chain, Shipping Optimization, Streamlit Dashboard, Predictive Analytics.

I. Introduction

Supply chain optimization plays an important role in reducing transportation costs and improving customer satisfaction. Nassau Candy Distributor currently assigns products to factories using predefined allocation rules. These static assignments often lead to inefficient shipping routes, higher logistics costs, and delayed deliveries.

This project proposes an intelligent recommendation system that predicts shipping performance and recommends better factory allocations using machine learning algorithms. The solution enables businesses to improve logistics efficiency while maintaining profitability.

II. Problem Statement

The existing factory allocation process suffers from several challenges:

- Static factory assignment rules
- Higher transportation cost
- Longer shipping lead time
- Reduced operational efficiency

- No predictive decision support

The objective is to build a machine learning model that recommends better factory allocation strategies based on historical data.

III. Objectives

The main objectives of this research are:

- Analyze historical shipping and sales data.
- Build predictive machine learning models.
- Compare multiple regression algorithms.
- Select the best-performing model.
- Develop an interactive Streamlit dashboard.
- Generate business recommendations for factory optimization.

IV. Dataset Description

The dataset contains operational records from Nassau Candy Distributor. Important attributes include:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Region
- Product Name
- Sales
- Units
- Cost
- Gross Profit

Additional factory mapping was performed based on product categories.

V. Methodology

The project follows these steps:

1) *Data Collection*: The Nassau Candy Distributor dataset was imported using Pandas.

2) *Data Preprocessing*:

- Removed missing values
- Feature encoding
- Feature scaling using StandardScaler
- Factory mapping
- Train-Test Split

3) *Machine Learning Models*: Three regression models were developed:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

4) *Model Evaluation*: Performance was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

- R^2 Score

5) *Feature Importance*: Random Forest feature importance was used to identify the most influential variables affecting predictions.

6) *Dashboard Development*: An interactive Streamlit dashboard was created to display:

- KPIs
 - Sales analysis
 - Factory-wise performance
 - Regional analysis
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- Monthly sales trends
 - Model comparison
 - Factory recommendation
 - Business insights

VI. Machine Learning Algorithms

A. Linear Regression

Linear Regression served as the baseline model by establishing a linear relationship between input variables and the target variable.

Advantages:

- Simple implementation
- Fast training
- Easy interpretation

B. Random Forest Regressor

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Advantages:

- High accuracy
- Handles nonlinear relationships
- Robust against overfitting
- Provides feature importance

C. Gradient Boosting Regressor

Gradient Boosting improves prediction performance by sequentially correcting previous errors.

Advantages:

- High predictive capability
- Better handling of complex datasets

VII. Experimental Results

The performance of all models was compared, as summarized in Table I.

Model	MAE	RMSE	R^2 Score
Linear Regression	Lowest baseline performance	Higher RMSE	Lower R^2
Random Forest	Best	Lowest RMSE	Highest R^2

Gradient Boosting	Good	Moderate RMSE	High R ²
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TABLE I
Model Performance Comparison

Random Forest achieved the best overall performance and was selected as the final recommendation model.

VIII. Streamlit Dashboard

A web-based dashboard was developed using Streamlit. Dashboard modules include:

- Dataset overview
- KPI cards
- Regional sales analysis
- Factory-wise sales visualization
- Monthly sales trend
- Correlation heatmap
- Model comparison
- Factory recommendation
- Business insights

The dashboard enables users to interactively analyze the dataset and simulate recommendations.

IX. Business Insights

Key observations include:

- Random Forest provided the highest prediction accuracy.
- Certain factories consistently generated higher sales.
- Regional demand significantly influences shipping efficiency.
- Product type impacts factory allocation decisions.
- Interactive dashboards support faster business decisions.

X. Conclusion

This project demonstrates how machine learning can improve factory allocation and shipping optimization for Nassau Candy Distributor. By comparing multiple regression algorithms, Random Forest was identified as the most accurate model. The Streamlit dashboard further enhances decision-making through interactive visualization and real-time recommendations. The proposed solution reduces logistics inefficiencies and provides valuable insights for supply chain optimization.

XI. Future Scope

Future improvements may include:

- Deep Learning models
- Real-time shipment tracking

- Geographic route optimization
- AI-powered recommendation engine
- Cloud deployment with live database integration

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